**Yu (Aaron) Shen Yu Shen**

# Probability Problem Solving in Data 140

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# Contents

Here are summaries of the core problem-solving methods from Data 140, explained with examples.

# 1
# Modeling (Construction/Deconstruction) of Randon Variable Distribution

Modeling is understanding. Modeling of random distribution is tell what a distribution is in terms of the available understandings. In this regard, modeling may also be called expression, expressing concept A in terms other concepts that have been understood, or better understood.

One of the often powerful expression is linear combination.

## 1.1 Randomness Characterizations

(We can't know everything of a randam variable's distribution, then we try to get some partial understanding of random distributions, as much as possible.)

*Mean* the weighted average of all possible values with the weights being the possibilities of the corresponding values. Showing the "center" of the value.

$E\left[X - E[X]\right] = 0$

*Variance* Variance measures the "spread" of a random variable around its mean ("center"). Mathematically, it is the expected squared deviation from the mean: (Why squared? indifferent to sign, want it to be non-negative) (Why not use other operation to get non-negative? e.g. absolute value $|X - E[X]|$? We want the transformation/operation to be continous, to be possible derivation, for example, for finding the maximum/minium of the variance ) (mathematically interesting: the difference between the mean of the square of $X$ and the square of the mean of $X$)

$Var(X) = E[(X - E[X])^2]$

$Var(X) = E(X^2) - (E[X])^2$
Simplification

*Standard deviation* the square root of the variance (Why perform square root operation on the variance? to be more comparable to the random variable's values, and its mean. )

$SD = \sqrt{(}Var(X))$

*Co-Variance*

*MGF*

## 1.2 Building blocks

### 1.2.1 Bernoulli distribution ($p$ probablity of success)

$P(X = k)$, $k \in \{1, 0\}$, $P(X = 1) = p$, then $P(X = 0) = 1 - p$, called $q$

- The simplest randonm distribution:
  - Only two outcomes (as one outcome wouldn't be random!)
- Express, relate between the outcome of 1 and 0, p to q.
- Mean: $E(X)$ : the average of all possible values. Only two possibilities, 1, 0, then the average of them: $1 \times p + 0 \times (1 - p)$ this evaluates to $p$

  Bernoulli $E[X] = p$
- Variance: $Var(X) = E(X^2) - (E[X])^2 = E(X^2) - p^2$ by simplified calculation

  Need to figure out $E(X^2)$ (we know $E(X)$ does not automatically implies that we know $E(X^2)$, it definitiely $E(X^2) \neq E(X)^2$)

  When there is no better means. Try the brutal force of calculating from the definition:

  $E(X^2) = \sum_{all\ possible\ values\ of\ X^2} p_{X^2} X^2$ where $p_{X^2}$ is the probability of a new random variable $Y$, $P(Y = X^2)$ as $X^2$ is a (1-to-1) function, thus $p_{X^2} = P(Y = k^2) = P(X = k) = p$ therefore, $E(X^2) = p1^2 + (1 - p)0^2 = p$ therefore, $Var(X) = E(X^2) - p^2 = p - p^2 = p(1 - p) = pq$

  $Bernoulli : E(X^2) = p$

  Bernoulli: $Var(X) = pq$

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### 1.2.2 Hypergeometric distribution

The Hypergeometric distribution models the number of successes $(g)$ in a fixed number of draws $(n)$ from a finite population $(N)$ when sampling is done without replacement.

### Key Characteristics Nature of Trials: Unlike the Binomial distribution, the trials are not independent because the probability of success changes with every draw as items are removed from the population. Parameters: It is defined by the population size $(N)$, the total number of successes available in that population $(G)$, and the number of items sampled $(n)$. Probability Mass Function (PMF):

$$P(X = g) = \frac{\binom{G}{g}\binom{N-G}{n-g}}{\binom{N}{n}}$$

where $g$ is the number of successes in your sample.

### Statistical Properties Mean: $n \cdot \frac{G}{N}$. Variance: $n \cdot \frac{G}{N} \cdot \frac{N-G}{N} \cdot \frac{N-n}{N-1}$. **Finite Population Correction:** The factor $\frac{N-n}{N-1}$ in the variance formula accounts for the reduction in variability caused by sampling a large portion of a finite population without replacement.

1. The deviravtion of hypergeometric

   The derivation of the Hypergeometric PMF is based on counting combinations when sampling without replacement from a finite population.

   (a) 1. The Scenario Suppose you have a population of size $N$ that contains $G$ "good" items (successes) and $B$ "bad" items (failures), where $N = G + B$. You take a simple random sample of size $n$ without replacement.

   The random variable $X$ represents the number of successes in your sample.

   (b) 2. The Combinatorial Logic To find $P(X = g)$, we use the "favorable outcomes over total outcomes" approach:

   *Total ways to choose the sample* There are $\binom{N}{n}$ ways to choose any $n$ items from the total population of $N$.

   *Ways to choose exactly $g$ successes* There are $\binom{G}{g}$ ways to pick $g$ successes from the $G$ available in the population.

   *Ways to choose the remaining failures* Since you picked $g$ successes, you must pick $n - g$ failures from the $B$ available failures. There are $\binom{B}{n-g}$ ways to do this.

   (c) 3. The Mass Function Multiplying the ways to get successes and failures gives the total favorable outcomes. Dividing by the total possible samples yields the PMF:

$$P(X = g) = \frac{\binom{G}{g}\binom{B}{n-g}}{\binom{N}{n}}$$

   This formula is valid for $0 \leq g \leq n$, provided $g \leq G$ and $n - g \leq B$.

   Does this combinatorial approach make sense, or would you like to see how this distribution's **expected value** $(n \cdot \frac{G}{N})$ is derived?

2. consecutive odds ratio of hypergeometric

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3. the difference between binomial and hypergeometric

   The main difference between the Binomial and Hypergeometric distributions lies in the independence of trials and the sampling method:

   *Binomial Distribution* Models the number of successes in $n$ independent trials where sampling is done with replacement.

- Because items are replaced, the probability of success remains constant for every trial.

*Hypergeometric Distribution* Models the number of successes in $n$ dependent trials where sampling is done without replacement from a finite population.

- The probability of success changes with each draw based on previous outcomes (not constant probability).

### Comparison of Properties

| Property | Binomial $(n, p)$ | Hypergeometric $(N, G, n)$ |
|---|---|---|
| :— | :— | :— |
| Independence | Independent trials | Dependent trials |
| Probability | Constant $(p)$ | Changes with each draw |
| Mean | $np$ | $n \cdot \frac{G}{N}$ |
| Variance | $np(1-p)$ | includes a finite population correction factor: $\frac{N-n}{N-1}$ |

When sampling a small fraction from a very large population, the dependence in the Hypergeometric distribution becomes negligible, and the Binomial distribution serves as a close approximation.

Does this help clarify why we use the "without replacement" distinction? Would you like to see how the Hypergeometric consecutive odds ratio is derived next? entered on: 2026-05-04 08:10:18

## 1.3 *Using better known distribution as building blocks to study other distributions*

The most classic example in Data 140 is the Binomial distribution, which is constructed entirely from Bernoulli trials.

### 1.3.1 *Building the Binomial Distribution*

A Binomial $(n, p)$, $n$ Bernoulli $(p)$ trials. (Repeated Bernoulli experiements)

The Component: Each trial $X_i$ is a Bernoulli random variable where $X_i = 1$ (success) or $X_i = 0$ (failure). The Construction: By adding them together

Binomial: $S_n = \sum_{i=1}^{n} X_i$

$X_i$ Bernoulli i.i.d.

you create a new distribution that counts the total number of successes in those $n$ trials. Why this works: This "building block" approach allows you to derive the properties of the Binomial distribution directly from the simpler Bernoulli trial. For instance, because expectation is linear, the mean of the Binomial is simply $n \times E[X] = np$.

Binomial $E[X] = np$

The Probability Mass (Density) Function (PMF) of binomial: – To derive the **Probability Mass Function (PMF)** for the Binomial$(n, p)$ distribution, we view it as a sequence of $n$ independent Bernoulli trials.

1. **1. Probability of a Single Specific Sequence** Imagine a specific sequence of $n$ trials that results in exactly $k$ successes and $n-k$ failures (e.g., $S, S, F, \ldots$). Because the trials are independent, we multiply their individual probabilities: The $k$ successes each have probability $p$, resulting in $p^k$. The $n-k$ failures each have probability $q=(1-p)$, resulting in $(1-p)^{n-k}$. The probability of that one specific sequence is $p^k(1-p)^{n-k}$.

2. **2. Counting All Possible Sequences** There are many different sequences that can result in $k$ successes. The number of ways to choose exactly $k$ positions for successes out of $n$ available trials is given by the binomial coefficient:

$$\binom{n}{k}=\frac{n!}{k!(n-k)!}$$

3. **3. The Final Density (PMF)** Since each of these $\binom{n}{k}$ sequences is mutually exclusive and has the same probability, we sum them to get the final formula:

$$P(X=k)=\binom{n}{k}p^k(1-p)^{n-k} \textbf{ for } k=0,1,\ldots,n$$

.

Binomial: $P(X=k)=\binom{n}{k}p^k(1-p)^{n-k}$ for $k=0,1,\ldots,n$ $\binom{n}{k}=\frac{n!}{k!(n-k)!}$

### 1.3.2 *Building Geometric Distribution*

You can also use Bernoulli trials to define "waiting time" distributions: The number of Bernoulli trials needed to get the first success. $P(X=k)=p(1-p)^{(k-1)}$ where $p$ is the probability of success.

Geometric distribution $(p)$: X $P(X=k)=p(1-p)^{(k-1)}$

### 1.3.3 *Building Negative Binomial Distribution*

The number of Bernoulli trials needed to reach a fixed number of successes.

To derive the Probability Mass Function (PMF) for the negative binomial distribution, we define the random variable $X$ as the total number of trials needed to achieve a fixed number of successes, $r$.

1. 1. The Logical Requirement For the $r$-th success to occur exactly on total trial $k$, two conditions must be met:
   (a) In the first $k-1$ trials, there must be exactly $r-1$ successes.
   (b) The $k$-th trial must be a success.

2. 2. Probability of the First $k-1$ Trials So the first part follows a Binomial distribution. We need to choose $r-1$ positions for successes out of $k-1$ available trials: Ways to arrange: $\binom{k-1}{r-1}$ Probability of $r-1$ successes: $p^{r-1}$ Probability of $(k-1)-(r-1)=k-r$ failures: $(1-p)^{k-r}$.

   This gives us: $\binom{k-1}{r-1}p^{r-1}(1-p)^{k-r}$.

3. 3. The Final Trial The $k$-th trial must be a success, which has probability $p$. Since the trials are independent, we multiply this by our previous result:

$$P(X = k) = \left[\binom{k-1}{r-1} p^{r-1}(1-p)^{k-r}\right] \times p$$

$$P(X = k) = \binom{k-1}{r-1} p^{r}(1-p)^{k-r} \text{ for } k = r, r+1, \ldots$$

.

Negative Binomial: $X$
$P(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}$
for $k = r, r+1, \ldots$
$k$: the number of total trials

4. Alternative Parameterization In some contexts, the distribution is defined by the number of failures $(k)$ observed before the $r$-th success. In that case, the total trials are $r + k$, and the PMF is written as:

$$P(X = k) = \binom{n+k-1}{k} p^{r}(1-p)^{k}$$

.

Two formulations of negative binomial (geometric):
1. ($k$: total number of trials to achieve fixed number $r$ success)
2. ($k$: number of failures to achieve fixed number $r$ success)
geometric: is negative binomial with $r = 1$ constant omitted

5. Why it is Negative Binomial? The Negative Binomial distribution is named that way primarily because it answers the "inverse" or "negative" version of the question posed by the Binomial distribution.

   Here are the two main reasons for the name:

   (a) Inverse Random Variables: In a Binomial distribution, you fix the number of trials and count the random number of successes. In a Negative Binomial distribution, you do the opposite: you fix the number of successes you want to achieve and count the random number of trials it takes to get there.

| Distribution | Fixed Quantity | Random Variable | Interpretation | Relationships |
|---|---|---|---|---|
| Binomial | # of trials $(n)$ | # of successes $(k)$ | Count successes in a fixed # of trials | $n \geq k$ |
| Negative Binomial | # of successes $(r)$ | # of trials $(k)$ | Count trials needed to reach a fixed # of successes | $k \geq r$ |
| Negative Binomial | # of successes $(r)$ | or # of failures $(k)$ | Count failures needed to reach a fixed # of successes | $k + r =$ total trials |

Both distributions assume independent Bernoulli trials with constant $p$

Practical modeling insight

Use Binomial when your experiment budget is fixed. Use Negative Binomial when your stopping condition is success-based (common in reliability testing, queuing, and retry systems).

If you want, I can map this to likelihood functions and show how they differ in inference workflows.

### 1.3.4 *Negative Binomial degenerates (collapses) to Geometric distribution*

To show that the Geometric distribution is a special case of the Negative Binomial distribution, we simply set the required number of successes $(r)$ to 1 in the Probability Mass Function (PMF).

1. The Mathematical Substitution

   (a) Negative Binomial PMF: This distribution models the number of trials $(k)$ to get $r$ successes:

   $$P(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}$$

   (b) Set $r = 1$:

   $$P(X = k) = \binom{k-1}{1-1} p^1 (1-p)^{k-1}$$

   (c) Simplify:

   - $\binom{k-1}{0} = 1$ (There is only one way to have the first success on the $k$-th trial: $k-1$ failures followed by one success).
   - Result: $P(X = k) = p(1-p)^{k-1}$.

   This final expression is exactly the PMF for the Geometric distribution, defined as the number of trials until the first success.

2. Conceptual Link

   - Negative Binomial: "How many trials until I get $r$ successes?".
   - Geometric: "How many trials until I get the first success?".

   When $r = 1$, the questions are identical.

### 1.3.5 *Law of Small Numbers: Poisson as approximation of the Binomial distribution*

"rare events" where the number of trials $(n)$ is large and the probability of success $(p)$ is very small, such that $np$ converges to a constant $\lambda$.

Detail deveolpment: The **Law of Small Numbers** describes how the Binomial distribution converges to the Poisson distribution when the number of trials $n$ is very large and the probability of success $p$ is very small, such that their product $np = \lambda$ remains constant.

Rare event of Binomial $X$: $\lim_{n\to\infty p\to 0} np = \lambda$

for Binomial $(n, p)$ $X$ $E[X] = np$

for Binomial $X_n$ $\lim_{n\to\infty} P(X_n = k) = \frac{\lambda^k e^{-\lambda}}{k!}$

1. **Mathematical Derivation** To derive this, we start with a variable $X_n \sim \text{Binomial}(n, \lambda/n)$ and take the limit as $n \to \infty$:

(a) Set up the Binomial PMF:

$$P(X_n = k) = \frac{n(n-1)\dots(n-k+1)}{k!}\left(\frac{\lambda}{n}\right)^k\left(1-\frac{\lambda}{n}\right)^{n-k}$$

(b) Rearrange the terms:

$$P(X_n = k) = \frac{\lambda^k}{k!}\left[\frac{n}{n}\cdot\frac{n-1}{n}\dots\frac{n-k+1}{n}\right]\left(1-\frac{\lambda}{n}\right)^n\left(1-\frac{\lambda}{n}\right)^{-k}$$

(c) Evaluate the limits as $n \to \infty$:

- The product in the brackets $[\frac{n}{n}\cdot\frac{n-1}{n}\dots\frac{n-k+1}{n}]$ consists of $k$ terms that each approach 1.
- The term $(1-\frac{\lambda}{n})^n$ approaches $e^{-\lambda}$.
- The term $(1-\frac{\lambda}{n})^{-k}$ approaches 1 because $k$ is a fixed constant while $n$ grows.

(d) Final Result: By combining these limits, we arrive at the Poisson PMF:

$$\lim_{n\to\infty} P(X_n = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

This approximation is considered accurate in practice when $n$ is large and the average $\lambda$ is relatively small.

2. Poisson Distribution Definition A random variable $X$ follows a Poisson $(\lambda)$ distribution if its Probability Mass Function (PMF) is:

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \text{ for } k = 0, 1, 2, \dots$$

where $\lambda > 0$ is the average rate of occurrence.

The Poisson distribution was developed as a mathematical limit of the **Binomial distribution**, discovered by Siméon Denis Poisson in 1837.

Poison $X$: $P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$ for $k = 0, 1, 2, \dots$

$\lambda > 0$ the avergae rate of occurrence

(a) **The Core Intuition: "The Law of Small Numbers"** The primary intuition source is the study of **rare events** occurring within a large number of independent trials. **The Limit:** If you have a sequence of trials $(n)$ that goes to infinity while the probability of success $(p)$ goes to zero, but their product $(np = \lambda)$ stays constant, the distribution converges to Poisson. **The "Building Block":** It essentially treats a continuous region of time or space as being divided into many tiny "bins" where a "success" (an event) is extremely unlikely in any single bin.

Poisson: rare events of arrival to some "bins" over (long) time or space with average rate of $\lambda$

(b) **Historical Real-World Sources** The distribution became famous through observations of phenomena that fit this "rare event" model perfectly: **Radioactivity:** Counting the clicks on a Geiger counter over time. **Military Data:** The number of Prussian soldiers killed by horse kicks annually across different army corps. **Wartime Clustering:** The hits from Nazi V1 "Buzz Bombs" in London during WWII, where the city was divided into small sectors to see if the bombs were clustering or falling randomly.

(c) **Key Logical Assumptions** For this intuition to hold, four things must be true:

i. **Independence:** One event does not affect the likelihood of another.

ii. **Constant Rate:** Events occur at a steady average frequency ($\lambda$).

iii. **Proportionality:** The number of events is proportional to the size of the interval.

iv. **Non-Simultaneity:** Two events cannot happen at the exact same instant or location.

### *1.3.6 Logarithmic Series distribution (not required in Data 140)*

The Logarithmic Series distribution is a discrete probability distribution used to model counts where the value zero is not possible, meaning its support is $k = 1, 2, 3, \ldots$.

1. Probability Mass Function (PMF) As a member of the **power series family**, its PMF is defined by the parameter $\theta$ $(0 < \theta < 1)$:

$$P(X = k) = \frac{\theta^k}{-k\ln(1-\theta)}$$

In this structure, the "coefficient" is $a(k) = 1/k$ and the "series function" is $f(\theta) = -\ln(1-\theta)$.

2. Meaning and Intuition

- Species Abundance: It is most famous for modeling biological diversity, such as the number of individual insects found for each species in a collection.
- Zero-Truncated Limit: Intuitively, you can think of it as the limiting form of a Negative Binomial distribution that has been "zero-truncated" (the $k = 0$ case is removed).
- Long-Tailed Behavior: It is a "long-tailed" distribution, meaning it can model scenarios where most categories have very few members, but a few rare categories have very high counts.

3. Visual Diagnosis In an Ord Plot, the Logarithmic Series is identified by a positive slope $(b > 0)$ and a negative intercept $(a < 0)$.

### *1.3.7 Power series family*

The **power series family** is a broad group of discrete probability distributions where the probability of a value $k$ is defined by the formula $P(X = k) = \frac{a(k)\theta^k}{f(\theta)}$.

In this structure, $\theta$ is a positive parameter, $a(k)$ is a coefficient function of $k$, and $f(\theta)$ is the "series function" that ensures the probabilities sum to one.

Power series family: $P(X = k) = \frac{a(k)\theta^k}{f(\theta)}$ , $\theta > 0$

1. Log-Ratio of Power Series Distributions Given $P(X=k)=\frac{a(k)\theta^k}{f(\theta)}$, then $R(k)=\frac{P(k)}{P(k-1)}=\frac{a(k)}{a(k-1)}\theta$ siginificantly simplified!

ratio of consecutive probabilities: $\frac{P(k)}{P(k-1)}=\frac{a(k)}{a(k-1)}\theta$

2. Components of Power Seriesu Distributions

| Distribution | $P(X=k)$ | Coefficient $(a(k))$ | Parameter $(\theta)$ | Series Function $(f(\theta))$ |
|---|---|---|---|---|
| Poisson ($k$: # of successes out of infinity) | $\frac{e^{-\lambda}\lambda^k}{k!}$ | $1/k!$ | $\lambda$ | $e^{\theta}$ |
| Binomial ($k$: # of successes of $n$ trials) | $\binom{n}{k}p^k(1-p)^{n-k}$ | $\binom{n}{k}$ | $p/(1-p)$ | $(1+\theta)^n$ |
| Negative Binomial ($k$: total # of trials to achieve fixed # $r$ success) | $\binom{k-1}{r-1}p^r(1-p)^{k-r}$ | $\binom{k-1}{r-1}$ | $1-p$ | $\left(\frac{\theta}{(1-\theta)}\right)^r$ |
| Negative Binomial ($k$: # of failures to achieve fixed # $r$ success) | $\binom{r+k-1}{r-1}p^r(1-p)^k$ | $\binom{r+k-1}{r-1}$ | $1-p$ | $(1-\theta)^{-r}$ |
| Geometric ($k$: # of total trials till success) | $p(1-p)^{(k-1)}$ | 1 | $1-p$ | $\frac{\theta}{(1-\theta)}$ |
| Geometric ($k$: # of failures till success) | $p(1-p)^{(k)}$ | 1 | $1-p$ | $(1-\theta)^{-1}$ |
| Logarithmic Series | $\frac{\theta^k}{-k\ln(1-\theta)}$ | $1/k$ | $\theta$ | $-\ln(1-\theta)$ |

More compontes of the power series distributions:

| Distribution | $R(k)=\frac{P(k)}{P(k-1)}=\frac{a(k)}{a(k-1)}\theta$ | Linear Recurrence $kR(k)=ak+b$ |
|---|---|---|
| Poisson ($k$: # of successes out of infinity) | $\frac{\theta}{k}=\frac{\lambda}{k}$ | $\lambda=0k+\lambda$ |
| Binomial ($k$: # of successes of $n$ trials) | $R(k)=\frac{n-k+1}{k}\cdot\frac{p}{1-p}$ | $\left(\frac{-p}{1-p}\right)k+\frac{p(n+1)}{1-p}$ |
| Negative Binomial ($k$: total # of trials to achieve fixed # $r$ success) | $\frac{k-1}{k-r}(1-p)$ | (hard to convert to $ak+b$ form, |
| | or let $k'=k-r$, $R(k')=\frac{k'+r-1}{k'}(1-p)$ | use the equivalent with $k$ as # of failures) |
| | In fact, $k'$ is just the # of failures! | $(1-p)k'+(r-1)(1-p)$ |
| Negative Binomial ($k$: # of failures to achieve fixed # $r$ success) | $R(k)=\frac{k+r-1}{k}\cdot(1-p)$ | $(1-p)k+(r-1)(1-p)$ |
| Geometric ($k$: # of total trials till success) | $\theta=1-p$ | $(1-p)k$ |
| Geometric ($k$: # of failures till success) | $\theta=1-p$ | $(1-p)k$ |
| Logarithmic Series | $\frac{k-1}{k}\theta$ | $\theta k-\theta$ |

3. Identifying the components of a power series distribution $P(X=k)=\frac{a(k)\theta^k}{f(\theta)}$:

   (a) Overall receipt:

      i. Identify $a(k)$: Isolate the coefficient function that depends on $k$ but is not part of the term with $k$ as an exponent.

      ii. Identify $\theta^k$: Group all terms where $k$ is the power; the base of this expression is your parameter $\theta$.

iii. Determine $f(\theta)$: The remaining terms (which do not depend on $k$) represent $\frac{1}{f(\theta)}$. Take their reciprocal to find the series function $f(\theta)$.

(b) Poisson In $P(X=k) = \frac{e^{-\lambda}\lambda^k}{k!}$:

i. $a(k) = 1/k!$ leaving $e^{-\lambda}\lambda^k$

ii. $\theta = \lambda$ leaving $e^{-\lambda}$

iii. $1/f(\theta) = e^{-\lambda} \Rightarrow f(\theta) = e^{\theta}$.

(c) Binomial

In $\binom{n}{k}p^k(1-p)^{n-k}$

i. $a(k) = \binom{n}{k}$ leaving $p^k(1-p)^{n-k}$ rearranging $p^k(1-p)^{n-k} = \left[\frac{p}{1-p}\right]^k (1-p)^n$

ii. $\theta = \frac{p}{1-p}$ leaving $(1-p)^n$

iii. $1/f(\theta) = (1-p)^n$ needing to express $1-p$ in terms of $\theta = \frac{p}{1-p}$

Trying to construct relationship between $1-p$ and $\theta$.

We have $p + (1-p) = 1$ creating the appearance of $\theta$ in the equation, by deviding $1-p$ at the both side of the above equation, resulting another equation:

$$\frac{p}{1-p} + 1 = \frac{1}{1-p}$$

thus we have $\theta + 1 = \frac{1}{1-p}$ therefore, we have $1-p = \frac{1}{\theta+1}$

thus

$$1/f(\theta) = (1-p)^n = \frac{1}{(\theta+1)^n}$$

Conclusion: $f(\theta) = (\theta+1)^n$

tricks: with $\theta = \frac{p}{1-p}$
use $p + (1-p) = 1$
to get $\frac{p}{1-p} + 1 = \frac{1}{1-p}$
$\theta + 1 = \frac{1}{1-p}$

(d) Negative binomial

In $\binom{k-1}{r-1}p^r(1-p)^{k-r}$ ($k$: total # of trials to achieve fixed # $r$ success)

i. $a(k) = \binom{k-1}{r-1}$ leaving $p^r(1-p)^{k-r}$ rearranging $p^r(1-p)^{k-r} = (1-p)^k p^r (1-p)^{-r} = (1-p)^k \left(\frac{p}{1-p}\right)^r$

ii. $\theta = 1-p$ leaving $\left(\frac{p}{1-p}\right)^r = \left(\frac{1-\theta}{\theta}\right)^r$

iii. $1/f(\theta) = \left(\frac{1-\theta}{\theta}\right)^r$ therefore $f(\theta) = \left(\frac{\theta}{1-\theta}\right)^r$

In $\binom{r+k-1}{r-1}p^r(1-p)^k$ ($k$: # of failures to achieve fixed # $r$ success)

i. $a(k) = \binom{r+k-1}{r-1}$ leaving $p^r(1-p)^k$

ii. $\theta = 1 - p$ leaving $p^r$ as $p = 1 - \theta$

iii. $1/f(\theta) = (1-\theta)^r$ therefore $f(\theta) = (1-\theta)^{-r}$

(e) Geometric

For $k$ to be the number of trials to get a success $k = 1, 2, \ldots$ In $p(1-p)^{(k-1)}$

i. $a(k) = 1$ remaining $p(1-p)^{(k-1)}$

ii. $\theta = 1 - p$ remaining $p(1-p)^{-1}$

iii. $\frac{1}{f(\theta)} = p(1-p)^{-1}$ expressing $p = 1 - \theta$, then $p(1-p)^{-1} = \frac{(1-\theta)}{\theta}$ therefore, $f(\theta) = \frac{\theta}{(1-\theta)}$

For $k$ to be the number of failurs before a success $k = 0, 1, 2, \ldots$ In $p(1-p)^{(k)}$

i. $a(k) = 1$ remaining $p(1-p)^{(k)}$

ii. $\theta = 1 - p$ remaining $p$

iii. $\frac{1}{f(\theta)} = p$ expressing $p = 1 - \theta$, then therefore, $f(\theta) = (1-\theta)^{-1}$

(f) Logarithmic Series

In $\frac{\theta^k}{-k\ln(1-\theta)}$

i. $a(k) = \frac{1}{k}$ leaving $\frac{\theta^k}{-\ln(1-\theta)}$

ii. $\theta = \theta$ leaving $\frac{1}{-\ln(1-\theta)}$

iii. $\frac{1}{f(\theta)} = \frac{1}{-\ln(1-\theta)}$ therefore, $f(\theta) = -\ln(1-\theta)$

4. Application of Power Series Distributions

(a) Mode of a distribution (distribution value of peak probability) The mode of a discrete distribution is a possible value of the distribution that has the highest probability. In a continuous distribution, it is the value where the probability density is at its peak. Key points to remember: Multiple Modes: A distribution can have more than one mode if multiple values share the highest probability. Binomial Mode: For a Binomial $(n, p)$ distribution, the mode is the integer part of $(n+1)p$. Identification: In Data 140, modes are often identified using consecutive odds ratios; as long as the ratio $R(k) \geq 1$, the distribution is rising or flat, and the point where it begins to fall identifies the mode.

mode of a distribution: distribution value(s) of peak probability

i. oods ratio (for finding mode) In the context of probability distributions like those studied in Data 140, an odds ratio (specifically a consecutive odds ratio) is the ratio of the probability of a value $k$ to the probability of the preceding value $k-1$,.
### Mathematical Definition For $k \geq 1$, the $k$-th consecutive odds ratio is defined as:

$$R(k) = \frac{P(k)}{P(k-1)}$$

,
### Why It Is Useful Simplified Computation: It allows you to calculate the entire distribution recursively starting from $P(0)$ without having to compute large factorials or powers for every value,. Determining Shape: Comparing $R(k)$ to 1 identifies the "direction" of the distribution's histogram,:

- $R(k) > 1$: The histogram is rising $(P(k) > P(k-1))$.
- $R(k) = 1$: The histogram is flat.
- $R(k) < 1$: The histogram is falling $(P(k) < P(k-1))$.

Finding the Mode: The largest $k$ for which $R(k) \geq 1$ identifies the mode (the most probable value) of the distribution,.
In categorical data analysis, the term is also used as a measure of association in $2 \times 2$ tables, comparing the odds of an event occurring in one group versus another.
Does this clarify how we use these ratios to "build" a distribution, or would you like to see the specific formula for the **Hypergeometric** odds ratio? entered on: 2026-05-04 07:58:23

(b) Usefulness of power series (abstraction) with its log-ratios The power series relationships are particularly useful because they allow us to linearize the probability mass function for graphical diagnosis.

i. 1. Odds Ratios vs. Log-Ratios

*Consecutive Odds Ratio* the raw ratio $R(k) = \frac{P(k)}{P(k-1)}$. It is used primarily for recursive calculation

- building the distribution step-by-step and
- finding the mode (where the ratio crosses 1).

*Log-Ratios* the logarithm of that ratio: $\log(R(k))$. Taking the log is the specific "trick" used for linearization.

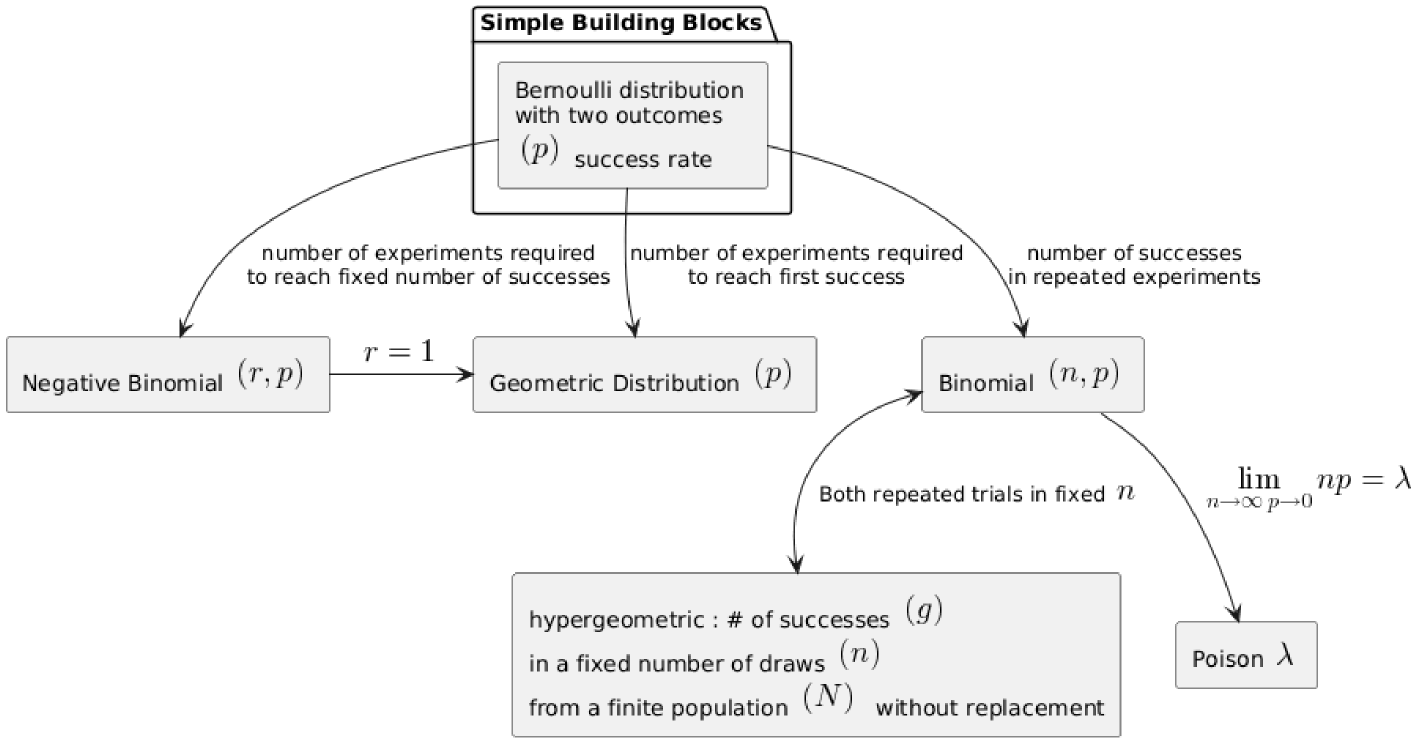
ii. 3. The Two Diagnostic Paths

A. Ord Plot: plots the relationship $k\frac{P(k)}{P(k-1)}(= ak + b)$. If the resulting plot is linear, the data belong to the Katz family (a subset of power series distribution). The specific distribution is identified by the signs of the slope $(a)$ and intercept $(b)$:

- Poisson: Slope is zero.
- Binomial: Slope is negative.
- Negative Binomial: Slope is positive and intercept is positive.
- Logarithmic Series: Slope is positive and intercept is negative.

B. Poissonness Plots: Plot a "count metameter" (which involves logs) against $k$. These are more robust because they use logs to isolate parameters like $\log(\lambda)$ as the slope.

### 1.3.8 "Family Tree" of Distributions



# 2
# Foundational Probability Calculations

## 2.1 Start from definitions

Perform reasoning and calculation definitions. This is often is the more straightforward approach that we don't use enough.

For example, the calculation of $E(X^2)$ where $X$ is Bernoulli

$E(X^2) = \sum_{all\ possible\ values\ of\ X^2} p_{X^2} X^2$

## 2.2 Function application to random variable

$P(X = k) = P(f(X) = f(x))$ When you apply a 1-to-1 (injective) function $f$ to a discrete random variable $X$, the probability of the new variable $Y = f(X)$ taking the value $f(k)$ is identical to the probability of $X$ taking the value $k$.

### 2.2.1 Why this works

Because the mapping is 1-to-1, the event $\{X = k\}$ is logically equivalent to the event $\{f(X) = f(k)\}$. Since they represent the same set of outcomes in the sample space, they must have the same probability:

$$P(X = k) = P(f(X) = f(k))$$

### 2.2.2 Examples from our discussion

1. Squaring a Bernoulli: We previously saw that for a Bernoulli variable $X \in \{0, 1\}$, applying $f(X) = X^2$ is a 1-to-1 mapping on its domain. Therefore, $P(X^2 = 1) = P(X = 1) = p$ and $P(X^2 = 0) = P(X = 0) = 1 - p$

2. Monotone Transformations: In Data 140, we often use monotone functions (which are always 1-to-1) to transform variables. a function is not 1-to-1 ($f(X) = X^2$ for a variable that can be both positive and negative) and how that changes the probability? For example, if $X \in \{-1, 0, 1\}$, then $f(X) \in \{0, 1\}$ (3 elements in input domain to 2 elements in the outputs, both 1 and -1 mapped to 1, that is, the probabily of $P(f(x) = 1) = P(X = -1) + P(X = 1)$)

## 2.3 *Linearity of Expectations:*

The expected value of a sum is the sum of the individual expectations, regardless of whether the variables are independent.

$$E\left[\sum_{i=1}^{n} \lambda_i X_i\right] = \sum_{i=1}^{n} \lambda_i E[X_i]$$

### 2.3.1 *Applicaiton Example: Why is $Var(X) = E(X^2) - (E[X])^2$ ?*

By definition: $Var(X) = E\left[(X - E[X])^2\right]$

By algebra expression (to figure out how to compute $(X - E[X])^2$ first, by $(a + b)^2 = (a - b)(a - b) = a^2 - 2ab - b^2$) $(X - E[X])^2 = X^2 - 2XE[X] + E[X]^2$

By linearity of expectation: $E\left[X^2 - 2XE[X] + E[X]^2\right] = E[X^2] - E\left[2XE[X]\right] + E\left[E[X]^2\right]$ For mean with respect to $X$, both $E[X]$ are constants, and $E[constant] = constant$ therefore $E\left[E[X]^2\right] = E[X]^2$ and $E\left[2XE[X]\right] = (2E[X])E[X] = 2E[X]^2$

By algeric substitution (replacing the equal value terms with simpler forms): $E\left[X^2 - 2XE[X] + E[X]^2\right] = E[X^2] - 2E[X]^2 + 2E[X]^2 = E[X^2] - E[X]^2$

### 2.3.2 *Applicaiton Example: The mean of a Binomial$(n, p)$ distribution is $np$ because it is the sum of $n$ Bernoulli$(p)$ trials, each with mean $p$.*

### 2.3.3 *Variance and Covariance Rules: The variance of a sum includes the sum of individual variances plus twice the covariances of all pairs.*

1. 1. Proof: Variance of a Sum The variance of a sum of random variables is derived from the definition of variance: $Var(X) = E[(X - E[X])^2]$

   For a sum $S = \sum_{i=1}^{n} X_i$:

   (a) Linearity of Expectation: $E[S] = \sum E[X_i]$.

(b) Substitution: $Var(S) = E\left[(\sum(X_i - E[X_i]))^2\right]$ and let $(X_i - E[X_i]) = a_i$, then $Var(S) = E\left[(\sum a_i)^2\right]$.

(c) Expansion: Expanding the square of a sum $(\sum a_i)^2$ gives $\sum a_i^2 + 2\sum_{i<j} a_i a_j$ (very elegant and convenient expression!), thus $Var(S) = E\left[\sum a_i^2 + 2\sum_{i<j} a_i a_j\right]$.

(d) Expectation of Terms:

- The terms $E[(X_i - E[X_i])^2]$ are the individual variances, $\sum Var(X_i)$.
- The cross-product terms $E[(X_i - E[X_i])(X_j - E[X_j])]$ are the covariances, $Cov(X_i, X_j)$.

Result: $Var(\sum X_i) = \sum Var(X_i) + 2\sum_{i<j} Cov(X_i, X_j)$.

If the variables are independent, all covariance terms are zero, simplifying the formula to $Var(\sum X_i) = \sum Var(X_i)$.

## 2.4 *MGF for Distribution Identification: The Moment Generating Function (MGF) of the sum of independent variables is the product of their individual MGFs.*

Example: Multiplying the MGFs of two independent Poisson variables ($\mu$ and $\lambda$) results in the MGF for a Poisson$(\mu + \lambda)$.

## 2.5 *Taylor Series Transformations: Using the expansion $e^z = \sum z^k / k!$ to simplify complex expectations into recognizable functional forms.*

Example: This was used to simplify the expectation $E[e^{tX}]$ to derive the Poisson MGF $e^{\lambda(e^t - 1)}$.

# 3

# *Variance and Co-Variance*

# 4

# *Optimization and Scaling*

1. **Log-Transformations:** Converting products into sums to simplify differentiation, which is essential for Maximum Likelihood (MLE) and MAP estimates [History].

   - **Example:** Taking the natural log of $x^6(1-x)^4$ to find the mode of a Beta distribution via simpler addition-based derivatives [History].

2. **Derivative-Based Optimization:** Setting the first derivative of a density or log-density to zero to find the parameter value that maximizes the probability [History].

   - **Example:** Differentiating the log-posterior $\frac{6}{x} - \frac{4}{1-x} = 0$ to find the MAP estimate of $0.6$ [History].

3. **Monotonic Function Application:** Since $\ln(x)$ is a monotonic increasing function, the value of $x$ that maximizes $\ln(f(x))$ is guaranteed to be the same value that maximizes $f(x)$ [History].

   - **Example:** This allows us to ignore complex constants in the Beta density when finding the MAP [History].

# 5
# *Discrete and Bayesian Inference*

1. **Recursive Odds Ratios:** Calculating $P(k)$ using the ratio $R(k) = \frac{P(k)}{P(k-1)}$, which avoids computationally heavy factorials for large $n$.
   - **Example:** To find $P(36)$ in a Binomial(60, 2/3), multiply $P(35)$ by the simple ratio $\frac{61-36}{36} \cdot 2$.
2. **Discrete Mode Identification:** Finding the peak of a histogram by identifying where the odds ratio $R(k)$ drops below 1.
   - **Example:** The mode of a Binomial$(n, p)$ is the integer part of $(n + 1)p$.
3. **Conjugate Prior Updating:** Using the "Beta-Binomial" partnership to update beliefs by simply adding successes and failures to the prior parameters [22, History].
   - **Example:** A Beta$(2, 3)$ prior combined with 2 heads in 7 tosses instantly becomes a Beta$(4, 8)$ posterior [History].
4. **Order Statistics Mapping:** (Poisson Approximation)** **Method:** When $n$ is very large and $p$ is very small, you can approximate Binomial probabilities using the **Poisson** $(\lambda)$ distribution, where $\lambda = np$. **Example:** Calculating the probability of a rare disease affecting exactly 5 people in a city of 100,000. Instead of a massive Binomial calculation, you use the Poisson PMF with $\lambda = 100,000 \cdot p$.

# 6

## 15. Poissonization

**Method:** A technique where you treat the total number of trials $n$ as a random variable following a Poisson distribution. **Why it works:** This often "decouples" variables that are normally dependent. For example, if the number of people entering a store is Poisson, the number of men and women entering become **independent** Poisson variables.

# 7

# *16. Markov Chain Steady-State Analysis*

**Method:** Modeling systems that transition between states where the next step depends only on the current state. You find the long-run "steady state" by solving the matrix equation $\pi P = \pi$. **Example:** Predicting the long-term market share of two competing brands based on monthly customer switching probabilities.

# *8*

# *17. Central Limit Theorem (CLT) Approximation*

**Method:** For a large sample size $n$, the distribution of the sample sum (or mean) is approximately **Normal**, regardless of the original distribution's shape. **Example:** Estimating the probability that the average height in a sample of 200 people is within a certain range using the $Z$-table.

# 9

## 18. Beta-Binomial Probability (*The "C-Constant" Formula*)

**Method:** Calculating the probability of $k$ successes in $n$ trials when the success probability $p$ is unknown and follows a Beta$(r, s)$ distribution. **Formula:** $P(S_n = k) = \binom{n}{k} \frac{C(r,s)}{C(r+k,s+n-k)}$ where $C$ is the normalizing constant of the Beta density.

# 19. Multiple Linear Regression (Matrix Form)

**Method:** Finding the best-fit parameters $\beta$ for a model with multiple predictors using the matrix identity $\hat{\beta} = (X^T X)^{-1} X^T Y$. **Example:** Predicting a house price $(Y)$ based on square footage, age, and location $(X_1, X_2, X_3)$ simultaneously.

# 11
# Other pending topics to discuss

## 11.1 Covariance: $Cov(X_i, X_j) = E[(X_i - E[X_i])(X_j - E[X_j])]$

## 11.2 Unbiased Estimator:

When working with sample data, we use the divisor $(n-1)$ in the formula $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$ to ensure the estimate of $\sigma^2$ is unbiased.

### 11.2.1 2. Variance vs. Sample Variance

While the proof above describes the theoretical Variance $(\sigma^2)$, statisticians often work with Sample Variance $(S^2)$.

| Feature | Variance $(\sigma^2)$ | Sample Variance $(S^2)$ |
|---|---|---|
| :— | :— | :— |
| Nature | A fixed parameter of a population or distribution. | A statistic calculated from a finite sample of data. |
| Calculation | $E[(X-\mu)^2]$ | $\frac{1}{n-1}\sum(X_i - \bar{X})^2$. |
| Goal | Describes the true underlying spread. | Estimates the true population variance. |

1. 3. Why use $(n-1)$? (Unbiased Estimation) The sample variance $S^2$ is an unbiased estimator of $\sigma^2$ because its expected value equals the true population variance: $E[S^2] = \sigma^2$.

   The "Bias" Problem: Using the population mean $\mu$ would allow a divisor of $n$. However, we use the sample mean $\bar{X}$, which is already "centered" in the data. This makes the deviations $(X_i - \bar{X})$ smaller on average than $(X_i - \mu)$. The Correction: Dividing by $n-1$ (Bessel's correction) perfectly compensates for this underestimation, ensuring the estimate is accurate on average.

## 11.3 Independent random variables

Two random variables $X$ and $Y$ are mathematically **independent** if the probability of any joint outcome is simply the product of their individual probabilities.

### 11.3.1 Mathematical Definitions

Discrete Variables: $X$ and $Y$ are independent if, for all possible values $x$ and $y$:

$$P(X = x, Y = y) = P(X = x) \cdot P(Y = y)$$

. Continuous Variables: Independence occurs when the joint probability density function (PDF) is the product of the individual marginal densities:

$$f_{X,Y}(x, y) = f_X(x) \cdot f_Y(y)$$

.

### 11.3.2 Key Properties in Data 140

1. **Expectation:** For independent variables, the expectation of their product equals the product of their expectations: $E[XY] = E[X]E[Y]$.

2. **Variance:** The variance of a sum of independent variables is simply the sum of their variances $(Var(X + Y) = Var(X) + Var(Y))$ because their **covariance is zero** [History].

3. **MGFs:** The Moment Generating Function of a sum of independent variables is the product of their individual MGFs: $M_{X+Y}(t) = M_X(t) \cdot M_Y(t)$ [History].

4. **Conditional Probability:** If $X$ and $Y$ are independent, knowing the value of $Y$ does not change the distribution of $X$: $P(X \mid Y) = P(X)$ [History].

In the **Bayesian framework** we discussed, we often move from assuming trials are "independent" to viewing them as **exchangeable**, where future trials depend on the evidence gathered from previous ones [History].

Would you like to see how this definition of independence is used to prove that the sum of independent **Normal** variables is also Normal?

# 12
# *Appendix: Hints of remembering the number of combinations formula*

$\binom{n}{k}$

Use the following equation:

$$n! = \binom{n}{k} k!(n-k)!$$

The total permualtions of $n$ elements can be achieved by all possible $k$ permutations and $(n-k)$ permutations. $n! = \binom{n}{k} k!(n-k)!$

# 13
# Core skills of mathematics required for probability

## 13.1 Calculus

1. Limit operation
2. Derivation
3. Integral
4. Function expression as series
   (a) Taylor series
5. $e$

## 13.2 Linear algebra

## 13.3 Combinatory

## 13.4 Trigonometry

# 14
# *If $\lambda$ is not an integer, the mode of a Poisson distribution is the **floor** (the "bottom") of $\lambda$, denoted as $\lfloor \lambda \rfloor$.*

### **Why?** The reason is found in the **consecutive odds ratio**, which tells us the "direction" of the distribution's shape:

$$R(k) = \frac{P(k)}{P(k-1)} = \frac{\lambda}{k}$$

**Rising:** As long as $k < \lambda$, the ratio $R(k) > 1$, meaning each probability is larger than the one before it. **Falling:** Once $k > \lambda$, the ratio $R(k) < 1$, and the probabilities begin to decrease.

If $\lambda$ is not an integer, the highest point reached before the ratio drops below 1 is the largest integer less than $\lambda$ (the floor). For example, if $\lambda = 2.7$, the probabilities rise until $k = 2$ and fall at $k = 3$, making **2** the mode.